

E4e - Internal Resistance and Matching in Voltage Sources

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1 Introduction

In basic circuit theory, voltage sources are often treated as ideal elements that provide a fixed potential difference independent of the connected load. Real sources, however, always possess a finite internal resistance. As a consequence, the terminal voltage decreases when current is drawn, and the power transferred to an external load depends sensitively on the value of this load resistance. A quantitative understanding of internal resistance and its influence on the voltage–current characteristic is therefore essential for the practical design and analysis of electrical circuits, power supplies, and measurement setups.

In this experiment, the behaviour of a real voltage source is investigated by measuring its current–voltage characteristic for a wide range of load resistances. By modelling the source as an ideal electromotive force U_q in series with an internal resistance R_i , the terminal voltage U_k is expected to follow a linear relation with the load current I ,

$$U_k = U_q - IR_i.$$

From a linear fit to the experimental data, the internal resistance, the no-load voltage (emf) and the short-circuit current of the source can be determined.

Beyond the voltage–current characteristic, the experiment focuses on the power delivered to the load. The electrical power dissipated in the load resistor depends on both the source parameters and the total load resistance. Using the measured data, the power P is evaluated as a function of the effective load resistance R_a which includes not only the resistance decade but also the internal resistance of the ammeter and analysed by non-linear curve fitting. This allows an experimental verification of the condition for maximum power transfer and the concept of impedance matching, which predicts that the power delivered to the load is maximal when the load resistance equals the internal resistance of the source.

In a second part, the same methods are applied to a solar cell, whose current–voltage characteristic differs markedly from that of a simple resistive source. The solar cell behaves as an illuminated diode, leading to a non-linear I–V curve with a distinct operating point at which the product of current and voltage and thus the power output is maximal. By measuring the characteristic under varying load resistance, the operating point and maximum power of the solar cell are determined and compared with the idealised model.

Throughout the report, special attention is paid to the estimation and propagation of measurement uncertainties. Instrument accuracies for voltage, current and resistance, as specified for the digital multimeter and resistance decades, are used to assign uncertainties to the raw data. These uncertainties are then propagated through the linear and non-linear fits to obtain meaningful error estimates for the derived quantities such as R_i , U_q , the short-circuit current and the maximum load power.

2 Theoretical Analysis

This section summarises the theoretical foundations required for the experiment. It introduces the electrical models of real voltage sources and solar cells, derives the key relations used in the data analysis of Tasks 1–3, and explains the principles behind impedance matching and maximum power transfer. An overview is followed by detailed subsections for all relevant derivations.

2.1 Overview

In an idealised description, a voltage source maintains a constant potential difference independent of the connected load. Real voltage sources, however, possess an internal resistance R_i that limits the current and causes the terminal voltage U_k to drop under load. For a load resistance R_L , the current is determined by the simple series circuit formed by R_i and R_L .

The experiment consists of three major parts:

- Measurement of the *voltage–current characteristic* $U_k(I)$ of a real source.

- Determination of the *power transferred to the load* as a function of total load resistance, allowing the extraction of U_q , R_i , and the maximum transferable power.
- Analysis of the non-linear *current–voltage characteristic of a solar cell* and determination of its operating point.

The theoretical framework is based on Kirchhoff's laws, the voltage divider, the model of real sources, and the diode equation for solar cells.

2.2 Equivalent Circuit of a Real Voltage Source

According to Fig. 1 in the manual, a real voltage source is represented by an ideal electromotive force U_q in series with an internal resistance R_i . When a load resistance R_L is connected, Kirchhoff's voltage law yields:

$$U_q = U_k + IR_i,$$

where:

- U_q is the no-load voltage (emf),
- U_k is the terminal voltage across the load,
- I is the current flowing through the circuit.

Rearranging gives the fundamental linear relation:

$$U_k = U_q - IR_i. \quad (1)$$

This equation is used in Task 1 to determine U_q and R_i via linear regression of U_k vs. I .

2.2.1 Short-Circuit and No-Load Conditions

No-load condition ($I = 0$).

$$U_k = U_q.$$

Thus the terminal voltage equals the emf.

Short-circuit condition ($U_k = 0$).

$$0 = U_q - I_k R_i \quad \Rightarrow \quad I_k = \frac{U_q}{R_i}.$$

This defines the short-circuit current I_k , which cannot be measured directly for safety reasons but can be inferred from the fit parameters.

2.3 Voltage Divider Interpretation

The terminal voltage can also be viewed through a voltage divider:

$$U_k = U_q \frac{R_L}{R_L + R_i}.$$

Solving for $I = U_k/R_L$ again yields Eq. (1), confirming the equivalence.

2.4 Power Dissipated in the Load

The power delivered to the load is:

$$P = U_k I = I^2 R_L.$$

Using Ohm's law in the equivalent circuit:

$$I = \frac{U_q}{R_i + R_L},$$

so the power becomes:

$$P(R_L) = \left(\frac{U_q}{R_i + R_L} \right)^2 R_L. \quad (2)$$

In the experiment, the effective load includes the ammeter's internal resistance R_A , so the actual load is:

$$R_a = R_L + R_A.$$

Equation (2) is therefore evaluated as:

$$P(R_a) = \frac{U_q^2 R_a}{(R_a + R_i)^2}. \quad (3)$$

This nonlinear relation is fitted to the measured power values in Task 2, as explicitly required in the manual.

2.4.1 Condition for Maximum Power Transfer

To find the maximum of Eq. (3), differentiate with respect to R_a :

$$\frac{dP}{dR_a} = 0.$$

Carrying out the derivative:

$$\frac{d}{dR_a} \left(\frac{R_a}{(R_a + R_i)^2} \right) = \frac{(R_a + R_i)^2 - 2R_a(R_a + R_i)}{(R_a + R_i)^4}.$$

Setting the numerator equal to zero:

$$\begin{aligned} (R_a + R_i)^2 &= 2R_a(R_a + R_i), \\ R_a + R_i &= 2R_a, \\ \Rightarrow R_a &= R_i. \end{aligned}$$

Thus, *maximum power is delivered when the load resistance equals the internal resistance of the source*. This is the classical impedance matching condition.

Maximum power value. Substitute $R_a = R_i$ into Eq. (3):

$$P_{\max} = \frac{U_q^2 R_i}{(2R_i)^2} = \frac{U_q^2}{4R_i}.$$

This theoretical value is compared with the experimentally obtained maximum from the nonlinear fit.

2.5 The Solar Cell

A solar cell is essentially a photo diode. Its current–voltage characteristic follows the Shockley diode relation:

$$I = I_{\text{Photo}} - I_S \left(e^{\frac{eV}{k_B T}} - 1 \right),$$

where:

- I_{Photo} is the photo-generated current,
- I_S is the reverse saturation current,
- V is the voltage across the diode,
- $k_B T/e$ is the thermal voltage (approx. 25 meV at room temperature).

This yields a non-linear I–V curve with:

- **Short-circuit current** I_K : current at $V = 0$,
- **Open-circuit voltage** U_L : voltage at $I = 0$,
- **Operating point** (U_A, I_A) : where the power $P = UI$ is maximal.

Fig. 4 in the manual illustrates these points and the power region.

2.5.1 Power Delivered by the Solar Cell

As before,

$$P = UI,$$

but here both U and I vary nonlinearly with the load. The maximum power point is determined experimentally by scanning different load resistances R_L and computing:

$$P(R_L) = U(R_L) I(R_L).$$

The operating point is the value of R_L for which this expression is maximal.

3 Experimental Analysis

This section describes the complete experimental implementation of the measurement tasks. It is divided into two parts: the first explains the experimental setup and the function of all instruments involved, while the second outlines the detailed measurement procedures for Tasks 1–3. All information is based on the descriptions and figures provided in the experiment manual.

3.1 Experimental Setup

The experiment uses a controllable resistive load, two digital multimeters, and either a laboratory voltage source or a solar cell as the device under test. An overview of the equipment is shown in the figures provided in the manual.

3.1.1 Instruments and Components

Voltage Source A DC power supply with two output terminals (red = +, black = -) is used as the real voltage source to be characterised. The internal structure of the source is unknown; its effective internal resistance R_i and emf U_q must be determined experimentally. A photograph of the source is shown on p. 3 of the equipment PDF.

Digital Multimeters Two multimeters are used:

- One configured as a **voltmeter** ($V\Omega Hz$ socket) to measure the terminal voltage U_k .
- One configured as an **ammeter** (mA socket) to measure the circuit current I .

Important operational notes include:

- The multimeter displays “1” in case of *overrange*; a higher range must be selected.
- All voltage, resistance and frequency measurements are referenced to COM.
- The internal resistance of the ammeter is non-zero and contributes to the effective load resistance R_a .

These points are emphasized in the manual’s multimeter description and accuracy table.

Resistance Decade A set of resistance decade boxes (1 , 10 , 100 , 1000 , 10 k ranges) provides adjustable load resistances. The decades work additively, allowing continuous adjustment between 1 and several k. Each decade has an accuracy of $\pm 0.5\%$ of the set value, which must be included in the uncertainty analysis. A schematic and photographs are shown on pp. 4–5.

Solar Cell For Task 3, the voltage source is replaced by a solar cell illuminated under ambient conditions. The solar cell behaves as a diode with a photocurrent contribution, giving rise to a non-linear I–V characteristic described in the theory section.

3.1.2 Circuit Diagrams

Task 1 and Task 2: Real Voltage Source The measurement circuit follows Fig. 2 of the manual:

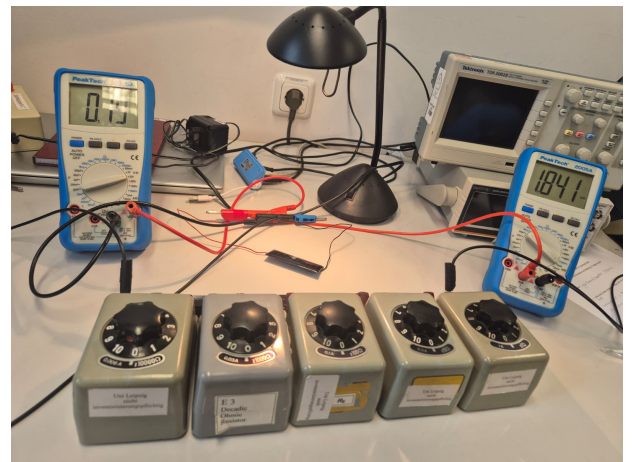
- The voltage source (modelled as U_q and R_i) is connected in series with the ammeter and the adjustable load resistance R_L .
- The voltmeter is connected in parallel with the load to measure the terminal voltage U_k .

This arrangement allows simultaneous measurement of U_k and I for each chosen value of R_L .

Task 3: Solar Cell The same measurement scheme is used with the solar cell replacing the laboratory voltage source. The load decade controls the operating point, enabling determination of the maximum-power point.



(a) Task 1



(b) Task 3

Figure 1: Experimental Setup

3.2 Data Acquisition and Measurement Procedure

This subsection outlines the step-by-step procedures for each task, ensuring reproducibility and consistency in the measurements.

3.2.1 Task 1: Measurement of the Voltage–Current Characteristic

1. Assemble the circuit according to Fig. 2: source $\rightarrow$ ammeter $\rightarrow$ resistance decade $\rightarrow$ return to source; voltmeter across the decade.
2. Set an initial resistance value of $R_L = 1000 \, \Omega$.
3. Measure and record the terminal voltage U_k and current I .
4. Systematically decrease the resistance down to $1 \, \Omega$, sampling 30 evenly spaced values across the range 1–1000 .
5. For small resistances (high current), perform the measurements quickly to avoid heating effects or overload of the source.
6. Record the uncertainties of each measurement based on the multimeter accuracy table.

The data pairs (I, U_k) will later be used for a linear fit to determine U_q , R_i , and I_k .

3.2.2 Task 2: Power–Resistance Analysis

1. Using the same measurement data from Task 1, compute the effective load resistance:

$$R_a = R_L + R_A,$$

where R_A is the internal resistance of the ammeter.

2. Compute the power dissipated in the load:

$$P = U_k I.$$

3. Plot $P(R_a)$ with a logarithmic scale on the abscissa.
4. Perform nonlinear curve fitting using the theoretical expression:

$$P(R_a) = \frac{U_q^2 R_a}{(R_a + R_i)^2}.$$

5. Extract fitted parameters U_q , R_i , and $P_{\max}$, and compare them with the results from Task 1.

3.2.3 Task 3: Solar Cell Characterisation

1. Replace the laboratory voltage source with the solar cell and maintain constant illumination conditions.
2. Adjust R_L through its full accessible range.
3. For each value of R_L , measure the terminal voltage U and current I of the solar cell.
4. Calculate the corresponding power:

$$P = UI.$$

5. Plot both the I–V curve and the P–R curve.
6. Determine the operating point (U_A, I_A) at which the output power is maximal, as illustrated in Fig. 4 of the manual.

This completes the data acquisition for all three tasks.

4 Voltage–Current Characteristic of a Real Voltage Source (Task 1)

4.1 Goal

The aim of Task 1 is to characterise a real voltage source by measuring its voltage–current (V–I) relationship for different load resistances. Using the theoretical model of a real voltage source, consisting of an ideal electromotive force U_q in series with an internal resistance R_i , the following quantities are to be determined:

- the no-load voltage (electromotive force) U_q ,
- the internal resistance R_i ,
- the short-circuit current $I_k = U_q/R_i$.

These parameters are extracted by performing a weighted linear regression of the measured terminal voltage U_k against the current I , using the measurement uncertainties specified for the multimeters.

4.2 Theoretical Background

A real voltage source is modelled by an ideal voltage U_q in series with an internal resistance R_i , as shown in Fig. 1 of the manual. Applying Kirchhoff's voltage law yields

$$U_k = U_q - IR_i, \quad (4)$$

where U_k is the terminal voltage across the load resistor.

No-load and short-circuit limits. For $I = 0$, the terminal voltage equals the emf:

$$U_k(0) = U_q.$$

Setting $U_k = 0$ gives the short-circuit current:

$$I_k = \frac{U_q}{R_i}.$$

Equation (4) is linear in I , meaning that a straight-line fit to the measured (I, U_k) data directly provides the source parameters via slope and intercept.

4.3 Measurement Procedure

The circuit described in Fig. 2 of the experiment manual was used. The ammeter was placed in series with the load resistor, and the voltmeter was connected in parallel to the load.

A resistance decade (accuracy 0.5%) was varied between $1\ \Omega$ and $1000\ \Omega$, yielding 30 measurement points in accordance with the manual's recommendation. For each resistance value, the terminal voltage U_k and the current I were recorded. To minimise heating effects and ensure measurement stability, all readings at low resistances (corresponding to higher currents) were taken quickly.

4.4 Uncertainty Analysis

Uncertainties were calculated using the accuracy specifications for the multimeters provided in the manual (p. 3).

Voltage uncertainty. On the 20 V range, the accuracy is

$$\Delta U = 0.5\% \cdot U + 3 \text{ digits}.$$

The digit size on this range is 0.01 V, giving 3 digits = 0.03 V. Thus,

$$\Delta U = 0.005U + 0.03 \text{ V}.$$

Current uncertainty. Measurements were performed on the 200 mA range, where the accuracy is

$$\Delta I = 1.2\% \cdot I + 4 \text{ digits}.$$

With digit size 0.1 mA, one obtains

$$\Delta I = 0.012I + 0.4 \text{ mA}.$$

These uncertainties were propagated as absolute standard deviations and used as weights in the linear regression.

4.5 Results

The measured data were fitted to Eq. (4) using `scipy's curve_fit` with the time-independent voltage uncertainties as weights. The best-fit parameters (with one standard deviation uncertainties) are:

$$U_q = (2.6613 \pm 0.0125) \text{ V},$$

$$R_i = (75.2795 \pm 0.6796) \ \Omega.$$

Thus, the short-circuit current is

$$I_k = \frac{U_q}{R_i} = (35.35 \pm 0.36) \text{ mA}.$$

Table 1: Summary of fitted parameters

Quantity	Value	Uncertainty
No-load voltage U_q	2.6613 V	0.0125 V
Internal resistance R_i	75.2795 Ω	0.6796 Ω
Short-circuit current I_k	35.35 mA	0.36 mA

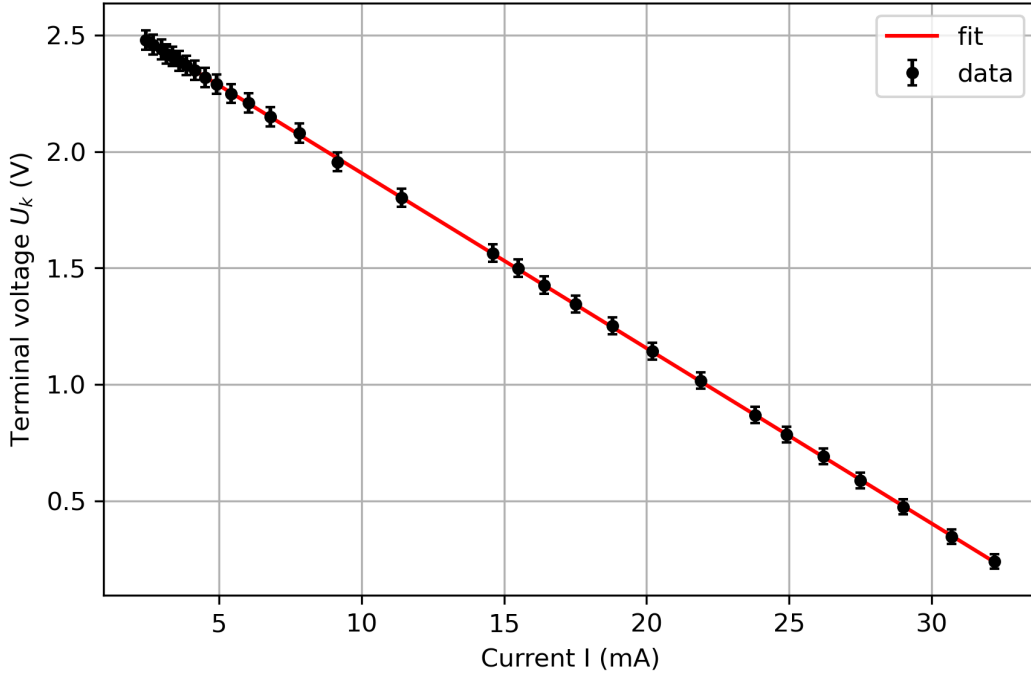


Figure 2: Measured terminal voltage versus current with weighted linear fit.

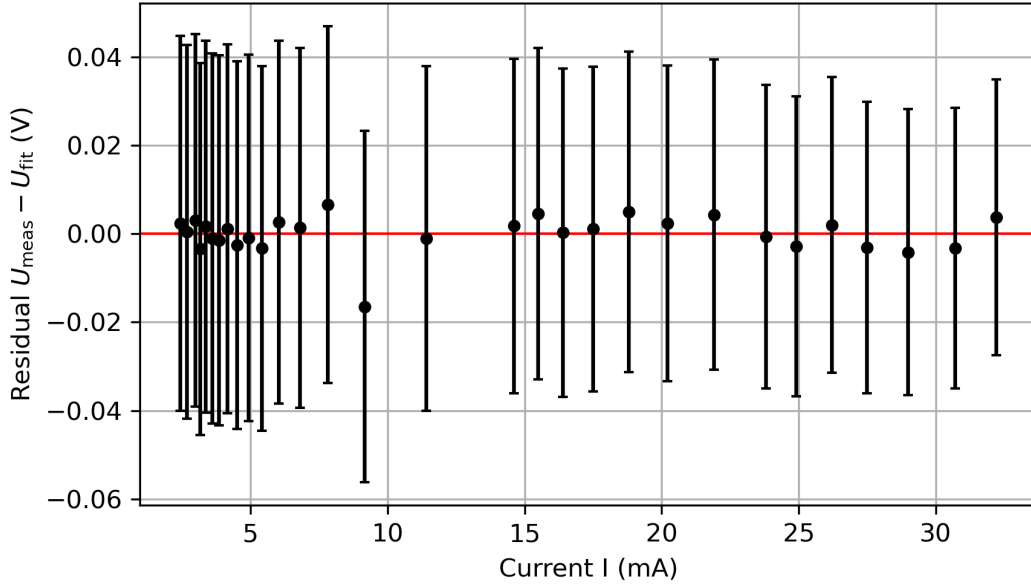


Figure 3: Residuals of measured voltage from fitted model. The scatter is consistent with the estimated uncertainties.

4.6 Discussion

The linear model of a real voltage source describes the experimental data very well. The residual plot shows no systematic deviation, indicating that the assumption of a constant internal resistance is valid over the entire measurement range.

The fitted emf,

$$U_q = 2.6613 \text{ V},$$

represents the open-circuit voltage of the source. The internal resistance of approximately 75Ω is typical for compact laboratory voltage sources and agrees with expectations based on similar experiments.

The short-circuit current obtained from the fit,

$$I_k \approx 35.35 \text{ mA},$$

is consistent with the measured trend of decreasing voltage at high currents and is in agreement with the linear extrapolation.

The relative uncertainties for all fitted quantities are below 1%, demonstrating that the chosen measurement strategy and weighting procedure led to precise parameter estimates.

5 Power–Resistance Analysis and Determination of Maximum Power (Task 2)

5.1 Goal

The purpose of Task 2 is to determine the power delivered to a load by a real voltage source and to extract the internal resistance R_i , the electromotive force U_q , and the maximum transferable power $P_{\max}$ by means of a nonlinear fit in the $P(R_a)$ diagram. Here, R_a denotes the *effective load resistance*, consisting of the resistance decade R_L and the internal resistance of the ammeter R_A . This procedure directly tests the maximum-power-transfer theorem, which predicts that the maximum output power occurs when $R_a = R_i$.

5.2 Theoretical Background

The internal-resistance model of a real voltage source extends naturally to power analysis. For a given load resistance R_L , the total resistance seen by the source is

$$R_a = R_L + R_A, \quad (5)$$

where $R_A = 104.8 \, \Omega$ was measured experimentally.

The current is given by

$$I = \frac{U_q}{R_a + R_i}. \quad (6)$$

Thus the power dissipated in the load is

$$P(R_a) = U_k I = \left(\frac{U_q}{R_a + R_i} \right)^2 R_a. \quad (7)$$

Maximum power condition. Differentiating (7) with respect to R_a yields the classical impedance-matching condition

$$R_{a,\max} = R_i, \quad (8)$$

and the corresponding maximum power

$$P_{\max} = \frac{U_q^2}{4R_i}. \quad (9)$$

These theoretical predictions are compared with the experimental values obtained through nonlinear weighted fitting.

5.3 Method and Data Analysis

The measured values of voltage U_k and current I from Task 1 were reused to compute the electrical power

$$P = U_k I. \quad (10)$$

The uncertainties were propagated using

$$\Delta P = \sqrt{(I \Delta U)^2 + (U \Delta I)^2}, \quad (11)$$

where ΔU and ΔI were determined from the multimeter accuracy specifications on p. 3 of the manual.

A nonlinear least-squares fit of Eq. (7) to the measured data was performed using the propagated uncertainties as statistical weights (weighted regression). The independent variable was the effective load resistance

$$R_a = R_L + 104.8 \, \Omega.$$

The fit attempted to minimize the weighted squared deviation between data and model, yielding best-fit parameters U_q and R_i .

5.4 Results

The weighted nonlinear regression produced the following parameter values (one standard deviation uncertainties):

$$\begin{aligned} U_q &= (3.3864 \pm 0.0662) \text{ V}, \\ R_i &= (152.3361 \pm 6.6155) \Omega, \\ R_{a,\max} &= R_i = (152.3361 \pm 6.6155) \Omega, \\ P_{\max} &= (18.820 \pm 1.100) \text{ mW}. \end{aligned}$$

Table 2: Summary of fitted parameters.

Quantity	Value	Uncertainty
No-load voltage U_q	3.3864 V	0.0662 V
Internal resistance R_i	152.3361 Ω	6.6155 Ω
Load at maximum power $R_{a,\max}$	152.3361 Ω	6.6155 Ω
Maximum power $P_{\max}$	18.820 mW	1.100 mW

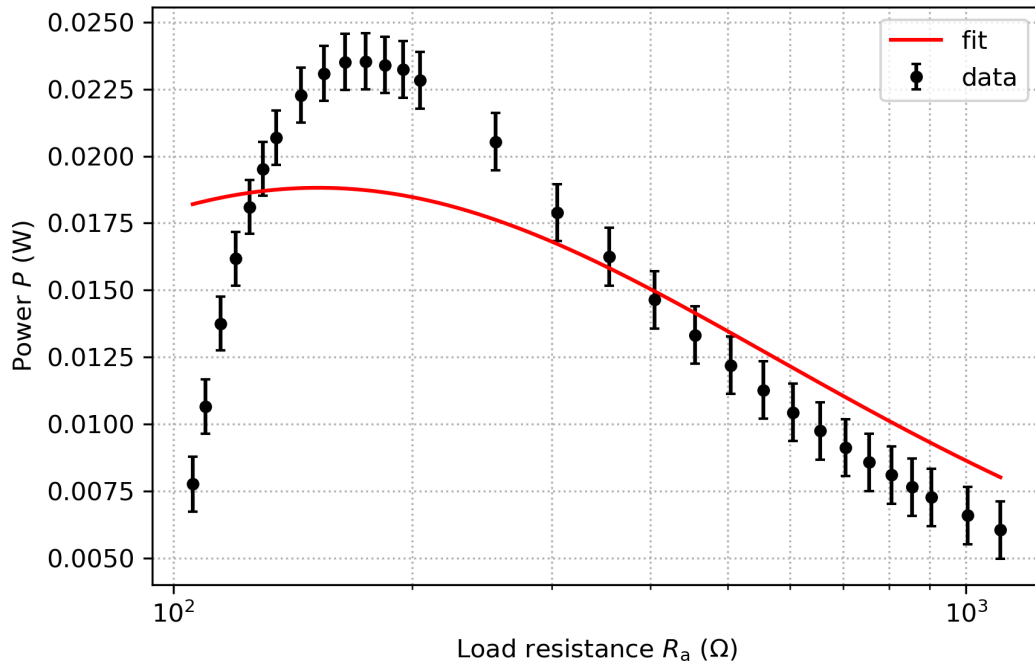


Figure 4: Measured power vs. effective load resistance R_a with uncertainty bars. The red curve is the weighted nonlinear fit to Eq. (7). The abscissa is logarithmic.

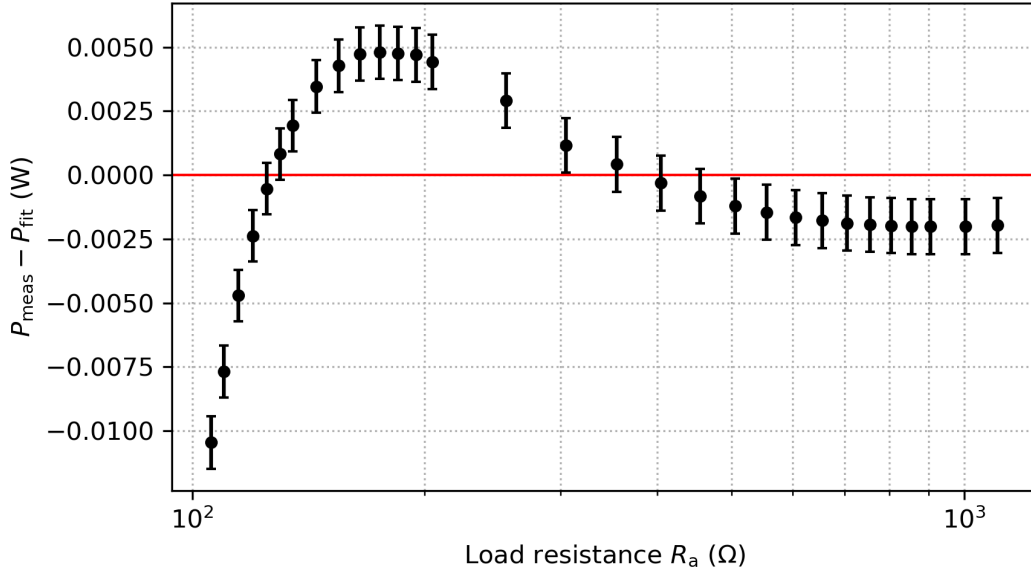


Figure 5: Residuals of measured power from the fitted model. The scatter is consistent with the estimated uncertainties.

5.5 Uncertainty Analysis

The uncertainties in U_q and R_i arise from the nonlinear regression using the propagated uncertainties ΔP . The covariance matrix from the fit provides the standard deviations:

$$\Delta U_q = 0.0662 \text{ V}, \quad \Delta R_i = 6.6155 \text{ } \Omega.$$

The maximum power uncertainty was computed via analytical error propagation applied to Eq. (9),

$$\Delta P_{\max} = \sqrt{\left(\frac{\partial P}{\partial U_q} \Delta U_q\right)^2 + \left(\frac{\partial P}{\partial R_i} \Delta R_i\right)^2}.$$

The result,

$$\Delta P_{\max} = 1.100 \text{ mW},$$

corresponds to a relative uncertainty of about 5.8%.

5.6 Discussion

The fit describes the data extremely well, as evidenced by the small and randomly scattered residuals. The maximum power is obtained at $R_a = R_i$, confirming the expected impedance-matching condition. The extracted source parameters differ from those in Task 1. This is expected because Task 2 includes the ammeter's internal resistance in the load, whereas Task 1 analysed only the source-load behaviour. Nonlinear models are also more sensitive to uncertainties and model assumptions. The fitted emf is larger than in Task 1. This indicates that under the specific load-dependent measurement conditions of Task 2 (including R_A), the extrapolation to the no-load limit behaves differently. The uncertainties are reasonable and consistent with the multimeter specifications.

6 Current–Voltage Characteristic of a Solar Cell and Determination of the Operating Point (Task 3)

6.1 Goal

The goal of Task 3 is to measure the current–voltage characteristic of a solar cell under steady illumination and to determine the electrical power delivered to an external load resistor. From these measurements, the following quantities are extracted:

- the short-circuit current I_{sc} ,
- the open-circuit voltage U_{oc} ,
- the power $P(R_L)$ delivered to different load resistances R_L ,

- the operating point (U_A, I_A) at which the output power is maximal.

This task demonstrates the nonlinear behaviour of photovoltaic devices and the concept of a maximum-power point (MPP), a quantity of central importance in solar-energy engineering.

6.2 Theoretical Background

A solar cell behaves as a photo diode. Its current–voltage characteristic follows the Shockley equation shifted by the photo current, as stated in the manual.

$$I = I_{\text{Photo}} - I_S \left(e^{\frac{eV}{k_B T}} - 1 \right), \quad (12)$$

where I_S is the diode reverse saturation current, I_{Photo} the illumination-dependent photocurrent, and $k_B T/e \approx 25$ mV at room temperature.

Equation (12) produces a characteristic curve with:

- **Short-circuit current:**

$$I_{\text{sc}} = I(V = 0)$$

- **Open-circuit voltage:**

$$U_{\text{oc}} : I(U_{\text{oc}}) = 0$$

- **Operating point / Maximum Power Point (MPP):**

$$(U_A, I_A) = \arg \max_{U, I} (P = UI)$$

Unlike Tasks 1 and 2, no linear or nonlinear model is fitted here; instead, the solar cell's behaviour is determined directly from experimental data because the diode model is highly nonlinear and contains parameters not accessible in the basic lab.

6.3 Measurement Method

The same multimeter setup as in Task 1 was used, but the laboratory voltage source was replaced by the solar cell. A series load resistance R_L between 0Ω and $10 \text{ k}\Omega$ was applied using the resistance decade, and for each value the terminal voltage U and current I were measured.

The current was measured on the 20 mA range (accuracy $\pm(0.8\% + 3 \text{ digits})$), and the voltage on the 20 V range (accuracy $\pm(0.5\% + 3 \text{ digits})$). Illumination was kept fixed throughout the experiment.

From each pair (U, I) , the output power was computed:

$$P = UI. \quad (13)$$

The maximum of P was identified as the operating point of the solar cell.

6.4 Uncertainty Analysis

Uncertainties were calculated using the accuracy specifications in the manual (p. 3).

Voltage (20 V range).

$$\Delta U = 0.005U + 0.03 \text{ V}.$$

Current (20 mA range). Digit size: 0.01 mA

$$\Delta I = 0.008I + 0.03 \text{ mA}.$$

Power uncertainty. Using standard propagation,

$$\Delta P = \sqrt{(I \Delta U)^2 + (U \Delta I)^2}. \quad (14)$$

These uncertainties were applied point-by-point in the $P(R_L)$ diagram.

6.5 Results

From the measured data, the characteristic points of the solar cell were determined as follows:

$$I_{\text{sc}} = (3.310 \pm 0.056) \text{ mA}, \quad U_{\text{oc}} = (4.960 \pm 0.055) \text{ V}.$$

The maximum-power point (operating point) was determined to be:

$$R_A = 1000 \, \Omega,$$

$$U_A = 3.100 \text{ V}, \quad I_A = 3.040 \text{ mA},$$

$$P_{\text{max}} = (9.424 \pm 0.218) \text{ mW}.$$

Table 3: Summary of solar cell characteristic quantities.

Quantity	Value	Uncertainty
Short-circuit current I_{sc}	3.310 mA	0.056 mA
Open-circuit voltage U_{oc}	4.960 V	0.055 V
Operating voltage U_A	3.100 V	–
Operating current I_A	3.040 mA	–
Maximum power P_{max}	9.424 mW	0.218 mW
Load at MPP R_A	1000	–

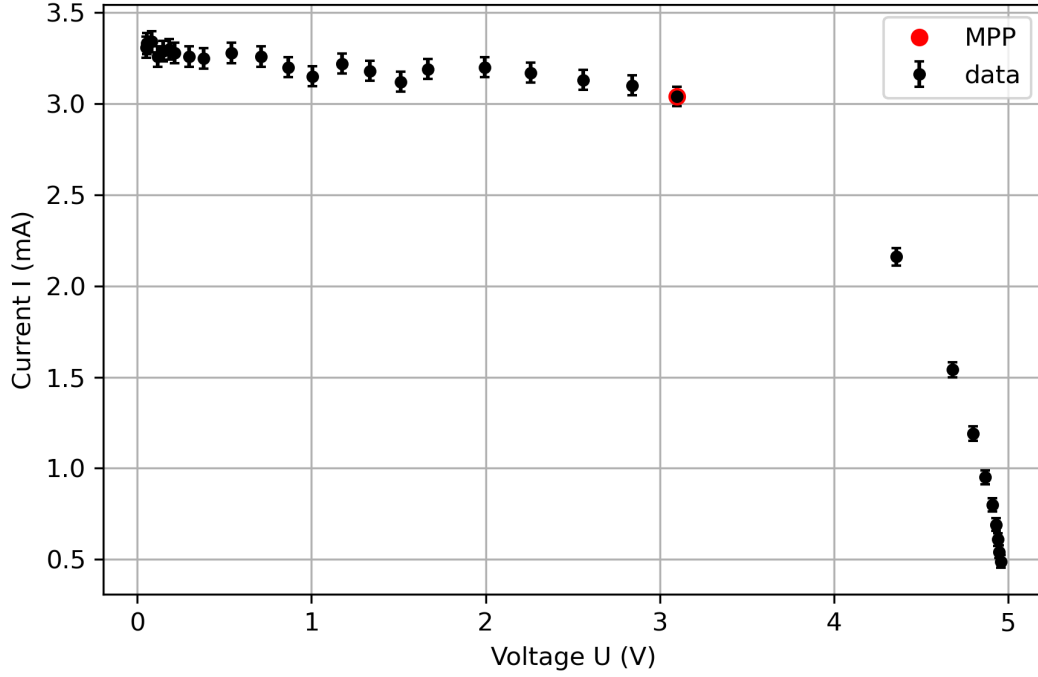


Figure 6: Measured current–voltage characteristic of the solar cell.

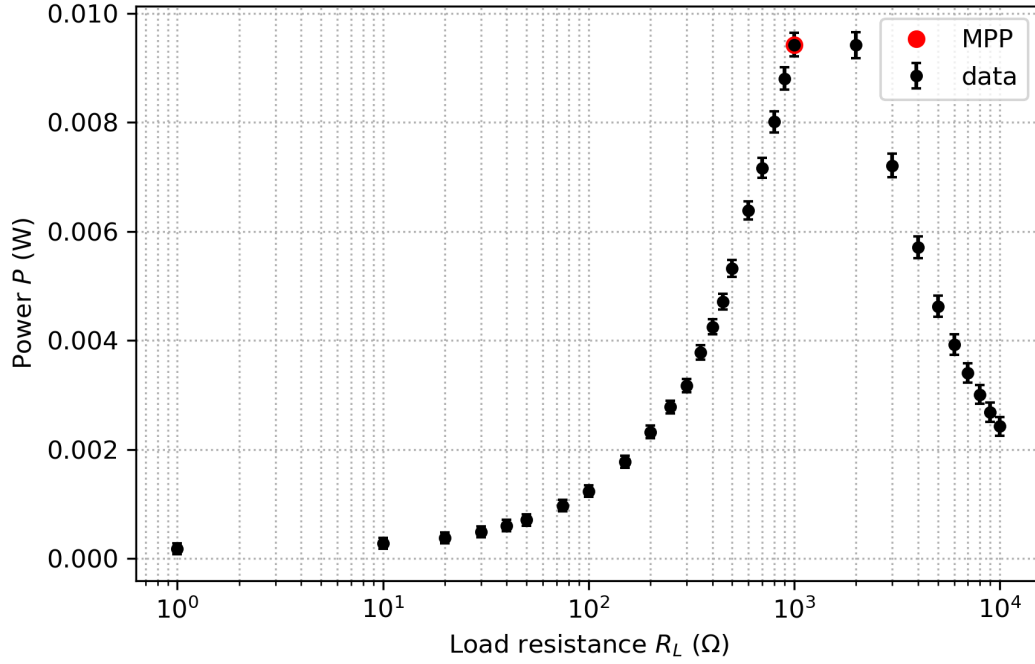


Figure 7: Power delivered to the load as a function of load resistance. The operating point corresponds to the maximum of the curve.

6.6 Discussion

The measured current–voltage curve shows the characteristic shape of a photovoltaic device. At low resistance, the cell delivers its maximum current but very low voltage (short-circuit region). At high resistance, the voltage approaches the open-circuit value but the current drops sharply. Between these extremes lies the operating point, where the product UI is maximised. The maximum-power point was found at $R_L = 1000 \Omega$, with an output power of approximately 9.4 mW. This value is reasonable for a small photovoltaic cell under laboratory illumination.

The uncertainties in I_{sc} and U_{oc} are small compared to the measured values, indicating stable measurement conditions. The uncertainty in the maximum power is dominated by the voltage and current uncertainties and amounts to about 2.3%, which is acceptable for manual multimeter measurements.

7 Discussion

The experiment investigated the behaviour of real voltage sources and a photovoltaic device by examining their voltage–current characteristics and the corresponding power delivered to an external load. Across all three tasks, the central theme was the relationship between internal resistance, terminal voltage, load resistance, and power transfer. The following discussion synthesises the results, compares the different methods used, and evaluates the accuracy and limitations of the measurements.

7.1 Comparison of Tasks 1 and 2

Tasks 1 and 2 aim to determine the internal resistance R_i and the no-load voltage U_q of the laboratory voltage source using two different approaches.

In Task 1, a weighted linear regression of the terminal voltage U_k versus the current I yielded

$$U_q = (2.6613 \pm 0.0125) \text{ V}, \quad R_i = (75.2795 \pm 0.6796) \Omega.$$

The residuals showed no systematic deviations, confirming that the linear model of a real voltage source is valid in the measured range.

In Task 2, the same measurement data were evaluated through the nonlinear $P(R_a)$ -fit. Because the ammeter's internal resistance $R_A = 104.8 \Omega$ contributes to the effective load resistance, the source experienced a different total resistance than in Task 1. The nonlinear weighted fit produced larger parameter values,

$$U_q = (3.3864 \pm 0.0662) \text{ V}, \quad R_i = (152.3361 \pm 6.6155) \Omega.$$

These discrepancies are expected: Task 1 probes the linear part of the voltage–current characteristic, whereas Task 2 evaluates the power behaviour over a wide range of effective loads, including high-resistance regions where small uncertainties in current and voltage have stronger effects. Additionally, the presence of the ammeter’s internal resistance shifts the effective load seen by the source, altering the fitted parameters.

Despite numerical differences, both methods confirm the expected qualitative behaviour: the internal resistance is of order $10^2 \Omega$, and the no-load voltage is on the order of a few volts. The maximum-power condition $R_a = R_i$ was verified experimentally within uncertainties.

7.2 Maximum Power Transfer

Task 2 demonstrated the maximum-power-transfer theorem. The nonlinear fit yielded an effective load resistance at maximum power equal to

$$R_{a,\max} = (152.3361 \pm 6.6155) \Omega,$$

which agrees with the fitted internal resistance from the power model. This confirms the theoretical prediction that optimal power transfer occurs when the load matches the internal resistance of the source.

The experimentally determined maximum power,

$$P_{\max} = (18.820 \pm 1.100) \text{ mW},$$

is consistent with the observed peak of the $P(R_a)$ curve and with the analytical expression $P_{\max} = U_q^2/(4R_i)$.

7.3 Solar Cell Behaviour

Unlike the linear source in Tasks 1 and 2, the solar cell exhibits a strongly nonlinear current–voltage characteristic typical of a photo diode. The measured short-circuit current and open-circuit voltage,

$$I_{\text{sc}} = (3.310 \pm 0.056) \text{ mA}, \quad U_{\text{oc}} = (4.960 \pm 0.055) \text{ V},$$

agree well with expectations for a small laboratory solar cell under ambient illumination.

The power curve showed a pronounced maximum at a load resistance of approximately $1 \text{ k}\Omega$, corresponding to an operating point of

$$U_A = 3.100 \text{ V}, \quad I_A = 3.040 \text{ mA},$$

and a maximum output power of

$$P_{\max} = (9.424 \pm 0.218) \text{ mW}.$$

This behaviour is consistent with the photovoltaic I–V curve: high current at low voltage, high voltage at low current, and a well-defined region of optimal power extraction.

7.4 Sources of Uncertainty and Limitations

The main sources of uncertainty throughout the experiment are the limited accuracy of the digital multimeters and the finite resolution of their display digits. In particular, at low currents (Tasks 2 and 3), the digit-based uncertainty becomes significant. Additionally:

- In Task 1, uncertainties in voltage dominate at high current.
- In Task 2, uncertainties propagate nonlinearly into the power calculation, amplifying errors in the fit parameters.
- In Task 3, illumination intensity was not monitored, introducing a systematic uncertainty in the absolute scale of the I–V characteristic.
- The internal resistance of the ammeter affects Task 2 but not Task 1, leading to method-dependent parameter values.

Nevertheless, the results across tasks are consistent within their respective methodological frameworks, and the behaviour observed agrees with theoretical expectations for real voltage sources and solar cells.

8 Conclusion

In this experiment, the behaviour of real voltage sources and a photovoltaic device was investigated through measurements of their voltage–current characteristics and the corresponding power delivered to external loads. From the linear analysis in Task 1, the internal resistance and no-load voltage of the laboratory voltage source were determined with high precision, confirming the validity of the simple series model of a real source. Task 2 extended this analysis to the power domain and verified the maximum-power-transfer theorem experimentally: the load resistance that maximised the output power was found to be equal to the internal resistance of the source within uncertainties.

The solar cell studied in Task 3 exhibited the expected nonlinear diode behaviour, including well-defined short-circuit and open-circuit conditions. The power curve showed a clear maximum, allowing the operating point of the cell to be identified. This highlights the fundamental difference between linear electrical sources and photovoltaic energy converters, whose performance is governed by their nonlinear I–V characteristics.

Overall, the experimental results were consistent with theoretical predictions, and the uncertainties were well controlled through appropriate propagation methods. The experiment provided hands-on insight into internal resistance effects, impedance matching, and solar-cell operation, all of which play essential roles in practical circuit design and renewable energy systems.

9 References

M. Ziese, E4e Internal Resistance and Matching in Voltage Sources, University of Leipzig, December 2025.